02451 Introduction to Machine Learning

Week 4: Probability theory, Bayes and Naive Bayes

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24 February 2026

DTU Compute, Technical University of Denmark

Today

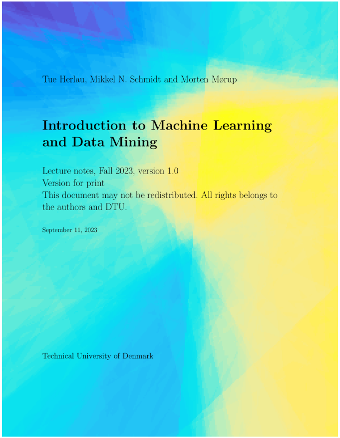
Feedback Groups of the day:

Chun-Po Chang, Qiju Chen, Tayshan Cole, Christian Alexander Dalgaard, Rasmus Ege, Bertram Feldborg, Rasmus Scheving Felsvang, Dagur Hall, Oliver Hvass, Selma Jakobsen, Anna Sky Kastl Jensen, Louise Kirkegaard, Søren Meldgaard Madsen, Nicklas William Lucas Madsen, Yana Malashkina, Antonios-Dimitrios Matanas, Alfred Leth Nielsen, Marius Drachmann Niss, Stefan Olevinskiy, Nikolaj Cervantes Ørskou, Marcus Piittala, Valdemar Højlund Rasmussen, Frank Rasmussen, Frida Robenhagen, Siri Roed Søndergaard, Alexander Adam Tribler.

Reading/homework material:

Chapter 5,(6),13

P5.1, P6.1, P6.2, P13.1, P13.2

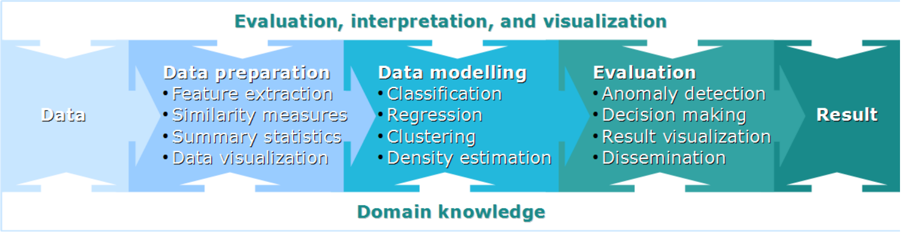


Lecture Schedule

1. Introduction, data, and visualization
   3 Febuary: C1,C2, C7
2. Summary statistics, similarity, and nearest neighbors
   10 Febuary: C4, C12
3. Computational linear algebra and PCA
   17 Febuary: C3
4. **Probability theory, Bayes and Naive Bayes**
   24 Febuary: C5, C6, C13 **(Assignment 1-3 due on 26 February)**
5. Bayesian inference and linear models
   3 March: C5, C6, C8
6. Decision trees, overfitting and cross-validation
   10 March: C9, C10
7. Performance evaluation and AUC
   17 March: C11, C16 **(Assignment 4-6 due on 19 March)**
8. Artificial Neural Networks and Optimization
   24 March: C15
9. Bias/Variance and ensemble methods
   7 April: C14, 17
10. K-means and hierarchical clustering
    14 April: C18 **(Assignment 7-9 due on 16 April)**
11. Mixture models and density estimation
    21 April: C19, C20
12. Representation learning
    28 April: Material provided separately
13. Recap and discussion of the exam
    5 May: C1-C20 **(Assignment 10-12 due on 7 May)**

Online help: Piazza
Videos of lectures: `https://panopto.dtu.dk`
Streaming of lectures: Zoom (link on DTU Learn)

Learning Objectives



Learning Objectives

- Understand basics of probability theory and stochastic variables.
- Understand probabilistic concepts such as Bayes theorem, expectations, independence.
- Understand `pmfs`, `pdfs` such as Bernoulli distribution and Gaussian distribution, and related concepts.
- Account for the assumptions made in Naïve Bayes.
- Apply Bayes theorem to obtain the class posterior.

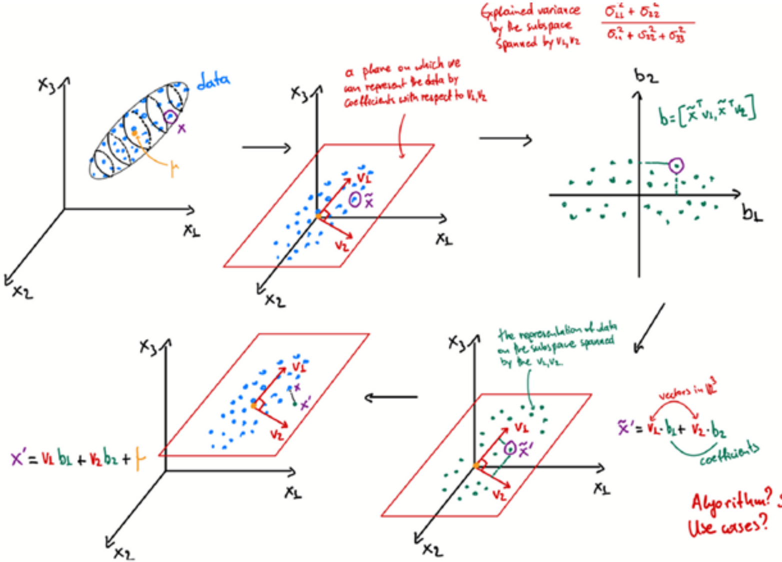
Plan for today:

- Lecture 4 (13:00 – $\sim$15:00)
  - **Discrete random variables and related concepts**
  - **Continuous random variables and related concepts**
  - **Bayes and Naive Bayes classifier**
- Exercises ($\sim$ 15:00–17:00)

Practicalities and announcements

- Assignment 04: Use (at least) version v20260223d
- Deadline for submitting assignments 1-3 this Thursday (26th February at 17:00). Remember to submit both HTML and ipynb!
- ... once submitted, please do the small survey on DTU Learn, indicating the time and perceived difficulty of the first batch of weekly assignments.
- TA session: Homework/exam at the end seemed to haev work well
- ...

Recap lecture 3



① $\tilde{X}$

② $\tilde{X} = U \Sigma V^T$

③ $b^T = V_{(k)}^T \tilde{x}$

Probabilities in machine learning

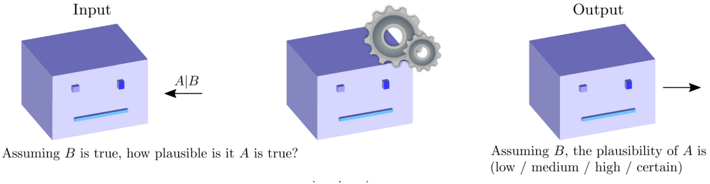
Pragmatically: A big part of ML/AI is dealing with **uncertainty** and **incomplete information**. Probabilities is the formal framework for doing so.

Algorithmically: **If** an image belongs to a particular category is a discrete event. The **probability** it belongs to a category is continuous. Algorithmically, easier to optimize continuous quantities.

Convenience: There are boiler-plate ideas for transforming a **probabilistic** assumption into an algorithm (maximum likelihood).

Probabilities

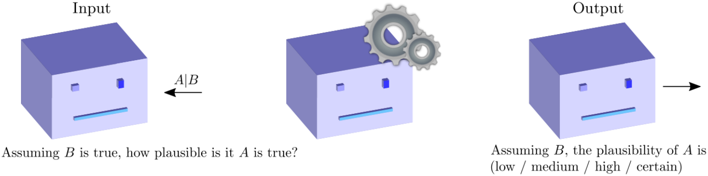
We reason about a proposition A in light of evidence B:



The degree-of-belief that A is true given B is accepted as true is at a level x

- A number between 0 and 1
- A and B are always binary (true/false) propositions
- Represents a *state of knowledge*

Probabilities: Trial example



$$
\begin{aligned}
G &: \textit{The accused is guilty} \\
E_1 &: \textit{A car similar to his was seen at the crime scene.} \\
E_2 &: \textit{A large sum of money was found in his posession} \\
E_3 &: \textit{His fingerprints was found at the door of the bank.}
\end{aligned}
$$

Probabilities express states-of-knowledge

$$E \equiv E_1 \text{ and } E_2 \text{ and } E_3$$

$$P(G|E) > P(G|E_2)$$

Binary propositions

A binary proposition is a statement which is either true or false (we might not know, but someone with complete knowledge would)

$$A : \textit{In 49 BCE, Caesar crossed the Rubicon}$$
$$B : \textit{Acceleration sensor 39 measures more than 0.85}$$
$$C : \textit{Patient 901 has high cholesterol}$$

Propositions can be combined with **and**, **or** and **not**:

$$AB \equiv \text{True if } A \text{ and } B \text{ are both true}$$
$$A + B \equiv \text{True if either } A \text{ or } B \text{ are true}$$
$$\overline{A} \equiv \text{True if } A \text{ is false}$$

We define two special propositions which is always **true**/**false**:

$$1 : \textit{A proposition which is always true}$$
$$0 : \textit{A proposition which is always false}$$

...and the following identities: $A1 = A, \quad A + \overline{A} = 1, \quad \overline{\overline{A}} = A$ and

$$A(B_1 + B_2 + \cdots + B_n) = AB_1 + AB_2 + \cdots + AB_n$$

Quiz 1: Probabilities

Assume we define the following 4 boolean variables.

$$
\begin{aligned}
R_1 &: \quad \textit{Handed in report 1} \\
R_2 &: \quad \textit{Handed in report 2} \\
R_3 &: \quad \textit{Handed in report 3} \\
F &: \quad \textit{Student failed 02450}
\end{aligned}
$$

How would you express the probability of the statement:

If a student hand in report 1, 2 and 3, the chance of passing 02450 is greater than 90%?

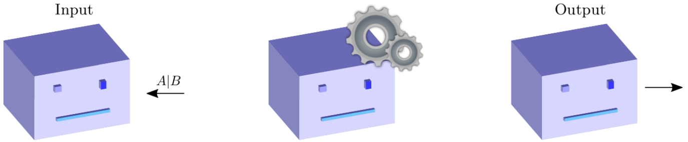
A. $P(R_1 R_2 R_3 | F) > 0.9$

B. $P(\overline{F} | R_1 + R_2 + R_3) > 0.9$

C. $P(\overline{F} | R_1 R_2 R_3) > 0.9$

D. $P(R_1 + R_2 + R_3 | F) > 0.9$

E. Don't know.



Assuming B is true, how plausible is it A is true?

Assuming B, the plausibility of A is (low / medium / high / certain)

Passing can be defined as not failing. Therefore, express the statement as:

$$P(\overline{F}|R_1R_2R_3) > 0.9$$

namely, if it is true report 1, 2 and 3 are all handed in, what is the chance of passing?

Rules of probability

$p(A) + p(\bar{A}) = 1$

The sum rule: $P(A|C) + P(\overline{A}|C) = 1$

The product rule: $P(AB|C) = P(B|AC)P(A|C)$

$= P(A|BC)P(B|C)$

Interpretation:

$P(A|B) = 0$ *(interpretation: given B is true, A is certainly false)*

$P(A|B) = 1$ *(interpretation: given B is true, A is certainly true)*

We also use the shorthand:

$$P(A|1) = P(A)$$

$$p(A) + P(\overline{A}) = 1$$

$$p(AB) = P(A|B)P(B)$$

$= p(B|A)p(A)$

Remarkably, this is the mathematical basis for this course

Marginalization and Bayes' theorem

Sum rule	$P(A\vert C) + P(\overline{A}\vert C) = 1$
Product rule	$P(AB\vert C) = P(B\vert AC)P(A\vert C)$

$= p(A|BC)p(B|C)$

$$
\begin{aligned}
P(B|C) &= P(B|C)\left[P(A|BC) + P(\overline{A}|BC)\right] = P(AB|C) + P(\overline{A}B|C) \\
&= P(B|AC)P(A|C) + P(B|\overline{A}C)P(\overline{A}|C).
\end{aligned}
$$

$$
\begin{aligned}
P(A|BC) &= \frac{P(B|AC)P(A|C)}{P(B|C)} \\
&= \frac{P(B|AC)P(A|C)}{P(B|AC)P(A|C) + P(B|\overline{A}C)P(\overline{A}|C)}.
\end{aligned}
$$

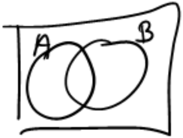
Exclusive and exhaustive events

A_1 : *The side ⚀ face up.* A_2 : *The side ⚁ face up.*

A_3 : *The side ⚂ face up.* A_4 : *The side ⚃ face up.*

A_5 : *The side ⚄ face up.* A_6 : *The side ⚅ face up.*

- When no two propositions can be true at the same time, they are said to be **mutually exclusive**: $A_i A_j = 0$ for $i \neq j$
- Consider any two events A and B



$$P(A+B) = P(A) + P(B) - P(AB)$$

- In general, for n mutually exclusive events

$$P(A_1 + A_2 + \cdots + A_n) = \sum_{i=1}^{n} P(A_i)$$

- A set of events is **exhaustive** if one has to be true: $A_1 + \cdots + A_n = 1$. Then:

$$\sum_{i=1}^{n} P(A_i) = P(A_1 + A_2 + \cdots + A_n) = 1$$

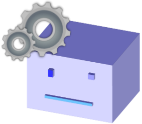
Exclusive and exhaustive events

A_1 : *The side* ⚀ *face up.* A_2 : *The side* ⚁ *face up.*

A_3 : *The side* ⚂ *face up.* A_4 : *The side* ⚃ *face up.*

A_5 : *The side* ⚄ *face up.* A_6 : *The side* ⚅ *face up.*

- When no two propositions can be true at the same time, they are said to be **mutually exclusive**: $A_i A_j = 0$ for $i \neq j$
- Consider any two events A and B



$$
\begin{aligned}
P(A+B) &= 1 - P(\overline{A}\,\overline{B}) & &= 1 - P(\overline{A}|\overline{B})P(\overline{B}) \\
&= 1 - \left[1 - P(A|\overline{B})\right] P(\overline{B}) & &= P(B) + P(A\overline{B}) \\
&= P(B) + P(\overline{B}|A)P(A) & &= P(B) + \left[1 - P(B|A)\right] P(A) \\
&= P(A) + P(B) - P(AB).
\end{aligned}
$$

- In general, for n mutually exclusive events $P(A_1 + A_2 + \cdots + A_n) = \sum_{i=1}^{n} P(A_i)$
- A set of events is **exhaustive** if one has to be true: $A_1 + \cdots + A_n = 1$. Then:

$$\sum_{i=1}^{n} P(A_i) = P(A_1 + A_2 + \cdots + A_n) = 1$$

Stochastic variables

- We often measure numerical quantities (number of children, age of a patient, etc.)
- Suppose a quantity X (number of children) takes a value $x = 3$. We can write this as the binary event X_3 and in general:

$$X_x : \{\text{The binary event that } X \text{ is equal to the number } x\}$$

- Stochastic variable simplify this notation by the definition:

$$X_x \longleftrightarrow X = x$$

$$P(X_x) \qquad P(X_x|Y_y) \longleftrightarrow P(X = x) = p(x) \qquad P(X = x|Y = y) = p(x|y)$$

Sum rule $P(A|C) + P(\overline{A}|C) = 1$

Product rule $P(AB|C) = P(B|AC)P(A|C)$

Marginalization

$$P(A|C) = P(A|BC)P(B|C) + P(A|\overline{B}C)P(\overline{B}|C)$$

Bayes theorem

$$P(A|BC) = \frac{P(B|AC)P(A|C)}{P(B|AC)P(A|C) + P(B|\overline{A}C)P(\overline{A}|C)}$$

$\longleftrightarrow$

Sum rule $\sum_i P(x_i|z_k) = 1$

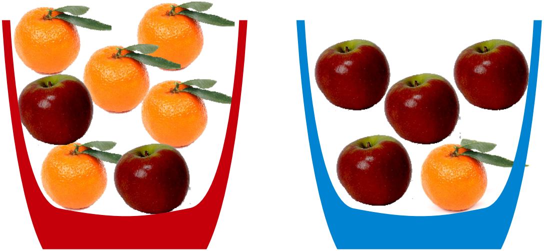
Product rule $p(x_i, y_j|z_k) = p(x_i|y_j, z_k)p(y_j|z_k)$

Marginalization $p(x_i|z_k) = \sum_j p(x_i|y_j, z_k)p(y_j|z_k)$

Bayes theorem $p(y_j|x_i, z_k) = \dfrac{p(x_i|y_j, z_k)p(y_j|z_k)}{\sum_j p(x_i|y_j, z_k)p(y_j|z_k)}$

Example: Everyday probabilities... are easy?

- What is the probability of an **orange** if the bowl is **red**?
- What is the probability of the **red** bowl if the fruit is **orange**?



Orange (clementine) taken from: `https://commons.wikimedia.org/wiki/File:Clementine_orange.jpg`
Apple taken from: `https://upload-wikimedia.org/wikipedia/commons/3/32/Dark_apple.png`

Probabilities

$p(\text{FruitType}, \text{Color})$

- Sum / marginalization rule

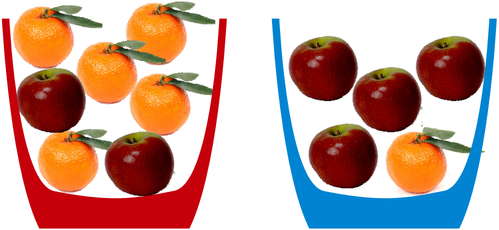
$$p(x) = p(x, y = 0) + p(x, y = 1)$$

- Product rule

$$\begin{aligned} p(x, y) &= p(x \mid y)p(y) \\ &= p(y \mid x)p(x) \end{aligned}$$

- Bayes rule

$$\begin{aligned} p(x \mid y) &= \frac{p(x, y)}{p(y)} \\ &= \frac{p(y \mid x)p(x)}{p(y)} \end{aligned}$$



14:10

- What is the probability of an **orange** if the bowl is **red**?
- What is the probability of the **red** bowl if the fruit is **orange**?

			
	2	5	7
	4	1	5
	6	6	12

$$p(o \mid r) = \frac{p(r,o)}{p(r)} = \frac{5/12}{7/12}$$

$$p(r) = p(r, o=0) + p(r, o=1)$$
$$= 2/12 + 5/12 = 7/12$$

$$p(r \mid o) = \frac{p(r,o)}{p(o)} = \frac{5/12}{6/12}$$

$$p(r \mid o) = \frac{p(o \mid r)p(r)}{p(o)}$$
$$= \frac{5/7 \times 7/12}{6/12}$$

14:10

Example: Everyday probabilities... can be tricky!

A medical test for a given disease:

- Correctly identifies the disease 99% of the time (true positives)
- Incorrectly turns out positive 2% of the time (false positives).

You know that

- 1% of the population suffers from the disease.
- As part of a random trial (uniform sampling) you are told to go to the doctor to get tested. The test turns out to be positive.

What is the probability you have the disease?

Hints:
We are looking for $p(Disease|Positive)$

Identify the following probabilities
$p(Positive \mid Disease)$
$p(Positive \mid NoDisease)$
$p(Disease)$
$p(NoDisease)$

Rules of probability:

$$p(y) = \sum_x p(x,y)$$
$$= p(y \mid x)p(x) + p(y \mid \neg x)p(\neg x)$$

$$p(x,y) = p(x \mid y)p(y)$$
$$p(x \mid y) = \frac{p(y|x)p(x)}{p(y)}$$

Example: Everyday probabilities... can be tricky!

- Correctly identifies the disease 99% of the time (true positives)
- Incorrectly turns out positive 2% of the time (false positives).
- 1% of the population suffers from the disease.

$$p(Disease) = \frac{1}{100} \qquad p(NoDisease) = \frac{99}{100}$$

$$p(Positive \mid Disease) = \frac{99}{100} \qquad p(Negative \mid Disease) = \frac{1}{100}$$

$$p(Positive \mid NoDisease) = \frac{2}{100} \qquad p(Negative \mid NoDisease) = \frac{98}{100}$$

$$\begin{aligned}
p(Disease \mid Positive) &= \frac{p(Positive \mid Disease)p(Disease)}{p(Positive)} \\
&= \frac{p(Positive \mid Disease)p(Disease)}{p(Positive, Disease) + p(Positive, NoDisease)} \\
&= \frac{p(Positive \mid Disease)p(Disease)}{p(Positive \mid Disease)p(Disease) + p(Positive \mid NoDisease)p(NoDisease)} \\
&= \frac{99/100 \times 1/100}{99/100 \times 1/100 + 2/100 \times 99/100} = \frac{99/100 \times 1/100}{99/100 \times 1/100 + 2/100 \times (1 - 1/100)} \\
&= \frac{\cancel{99/100} \times 1/\cancel{100}}{\cancel{99/100} \times 1/\cancel{100} + 2/\cancel{100} \times \cancel{99/100}} = \frac{1}{3}
\end{aligned}$$

Quiz 2: Probabilities

Consider a dataset which describe the consumption of delicatessen products in different cities. Each observation in the dataset is a customer, and we record the city the customer is from as well as their consumption of delicatessen. Suppose you are told:

- 17.5 % were from Lisbon, 10.7 % were from Oporto and 71.8 % from the Other region.
- 44.1 % of the costumers from Lisbon spent above the median consumption on delicatessen (DELI).
- 48.9 % of the costumers from Oporto spent above the median consumption on delicatessen (DELI).
- 51.6 % of the costumers from the Other region spent above the median consumption on delicatessen (DELI).

What is the probability based on the wholesale data that a costumer that spent above the median consumption on delicatessen (DELI) come from Lisbon?

A. 7.7 %

B. 15.4 %

C. 44.1 %

D. 59.6 %

E. Don't know.

We will let $LISBON$, $OPORTO$ and $OTHER$ denote coming from Lisbon, Oporto and the other region respectively. $DELI_H$ will denote above the median value of delicatessen consumption. We now find using Bayes theorem:

$$\begin{aligned}
P(LISBON|DELI_H) &= \frac{P(DELI_H|LISBON)P(LISBON)}{P(DELI_H)} \\
&= \frac{P(DELI_H|LISBON)P(LISBON)}{\begin{array}{c}P(DELI_H|LISBON)P(LISBON)\\ +P(DELI_H|OPORTO)P(OPORTO)\\ +P(DELI_H|OTHER)P(OTHER)\end{array}} \\
&= \frac{0.441 \cdot 0.175}{0.441 \cdot 0.175 + 0.489 \cdot 0.107 + 0.516 \cdot 0.718} \\
&= 0.1544 \approx 15.4\%
\end{aligned}$$

Independence

Independent: $p(x_i, y_j) = p(x_i)p(y_j)$

Conditionally independent given z_k : $p(x_i, y_j | z_k) = p(x_i | z_k)p(y_j | z_k)$

Expectations

$\mathbb{E}_{x \sim p(x)} \{f(x)\}$

$$\textit{Expectation:} \quad \mathbb{E}[f] = \sum_{i=1}^{N} f(x_i) p(x_i).$$

$$\text{mean: } \mathbb{E}[x] = \sum_{i=1}^{N} x_i p(x_i), \quad \text{Variance: } \mathrm{Var}[x] = \sum_{i=1}^{N} (x_i - \mathbb{E}[x])^2 p(x_i).$$

Example: Uniform probability

1, 2, 3, 4, 5, 6

$$p(x_i) = \frac{1}{N}$$

$$\mathbb{E}[f] = \frac{1}{N} \sum_{i=1}^{N} f(x_i)$$

$$\mathbb{E}[x] = \frac{\sum_{i=1}^{N} x_i}{N}$$

$$\mathrm{Var}[x] = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mathbb{E}[x])^2$$

The Bernoulli distribution

- Let $b = 0, 1$ denote a binary event.
- For instance,
  - $b = 0$ corresponds to heads, and $b = 1$ to tails, or
  - $b = 0$ corresponds to a person being ill, and $b = 1$ that a person is well.
- The probability of b is expressed using a parameter θ in the unit interval $[0, 1]$

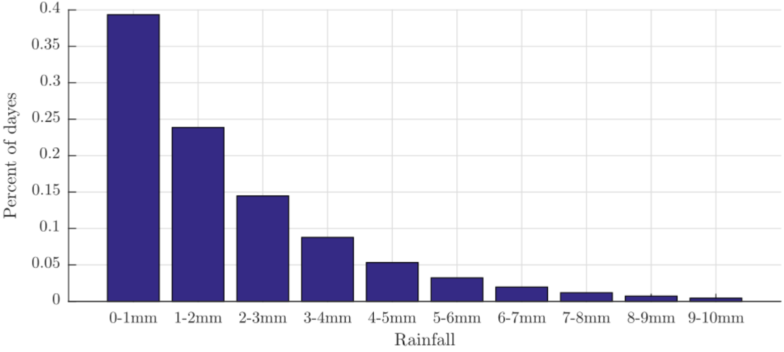
$$\textit{Bernoulli distribution:} \quad \underline{p(b|\theta)} = \theta^b (1-\theta)^{1-b}.$$

$b=1: p(b=1|\theta) = \theta$

$b=0: p(b=0|\theta) = 1-\theta$

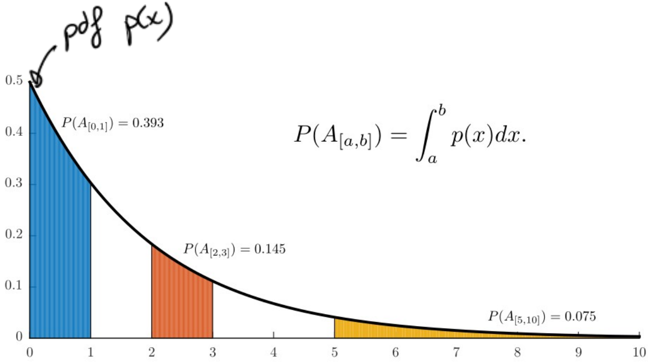
Probability vs. Density

- Suppose we consider the rainfall on an average day r
- **Can not** talk about the probability there will be **exactly** $r = 2.3$ mm of rain, $P(r = 2.3)$ mm
- **Can** talk about the probability there will be **between** 1 and 2 mm of rain



Probability vs. Density

- These probabilities can be **represented** as integrals
- **Events** are **intervals**, the **probability** is the **integral**, the curve is the **density**

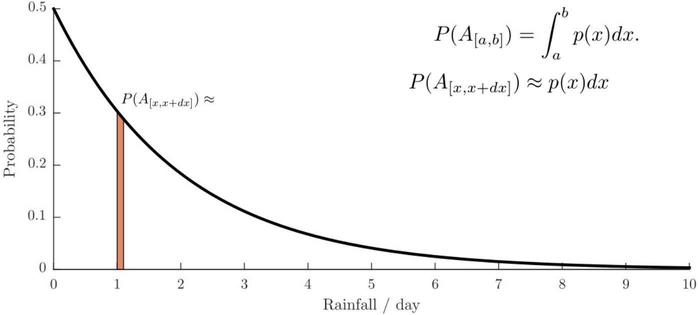


$$P(A_{[a,b]}) = \int_a^b p(x)dx.$$

$A_{[a,b]}$: *There will be between a and b mm of rain*

Probability vs. Density

- These probabilities can be **represented** as integrals
- **Events** are **intervals**, the **probability** is the **integral**, the curve is the **density**
- What is the probability there will be between 1 and 1.1 mm of rain?



$A_{[a,b]}$: *There will be between a and b mm of rain*

Probability vs. Density

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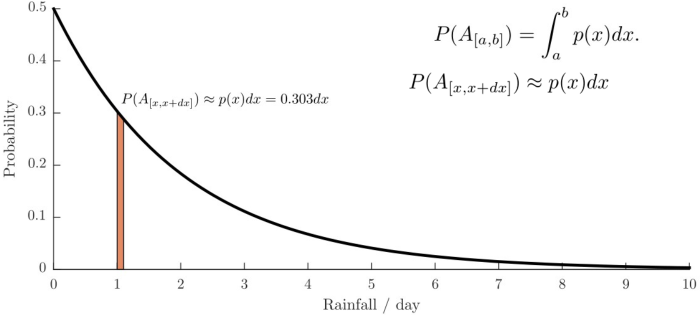


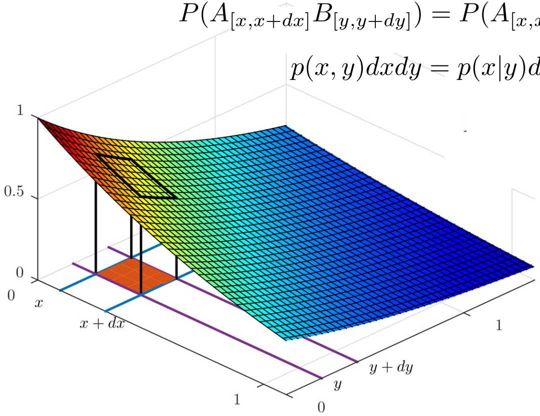
$A_{[a,b]}$: *There will be between a and b mm of rain*

Probability vs. Density

$$P((x,y) \in D) = \int_{(x,y)\in D} p(x,y)dxdy$$

$$P(A_{[x,x+dx]}B_{[y,y+dy]}) = P(A_{[x,x+dx]}|B_{[y,y+dy]})P(B_{[y,y+dy]})$$

$$p(x,y)dxdy = p(x|y)dxp(y)dy$$



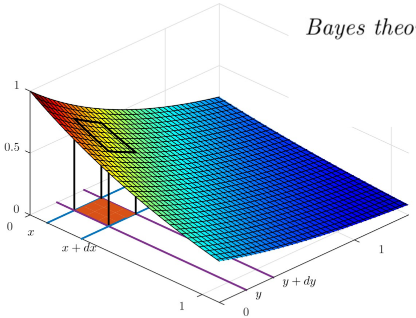
This implies:

$$p(x,y) = p(y|x)p(x)$$

Probability vs. Density

The sum rule: $\int dx \; p(x|z) = 1$

The product rule: $p(x, y|z) = p(y|x, z)p(x|z)$



Bayes theorem:

$$p(x|y, z) = \frac{p(y|x, z)p(x|z)}{p(y|z)} = \frac{p(y|z)p(x|y, z)}{\int p(y|x', z)p(x'|z)dx'}.$$

Collecting all of this we obtain:

For **discrete** random variables and probability mass functions:

Marginalization $$\sum_c p(x = c, y \mid z) = p(y \mid z)$$

Product rule $$p(x, y \mid z) = p(y \mid z, x)p(x \mid z)$$

Bayes theorem $$p(x \mid y, z) = \frac{p(y \mid z, x)p(x \mid z)}{\sum_c p(y \mid x = c, z)p(x = c \mid z)}$$

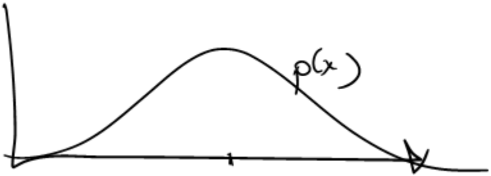
For **continuous** random variables and densities

Marginalization $$\int p(x, y \mid z)dx = p(y \mid z)$$

Product rule $$p(x, y \mid z) = p(y \mid z, x)p(x \mid z)$$

Bayes theorem $$p(x \mid y, z) = \frac{p(y \mid z, x)p(x \mid z)}{\int p(y \mid z, x')p(x' \mid z)dx'}$$

Expected value



- **Discrete** random variables

$$\mathbb{E}[g] = \sum_i g(x_i)p(x_i)$$

- **Countinous** random variables

$$\mathbb{E}[g] = \int g(x)p(x)\,dx$$

Expected value

- Mean

$$\bar{x} = \mathbb{E}[x] = \int x\, p(x)\, dx$$

- Covarinace

$$\begin{aligned} cov(x, y) &= \mathbb{E}[(x - \bar{x})(y - \bar{y})] \\ &= \mathbb{E}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])] \end{aligned}$$

- Variance

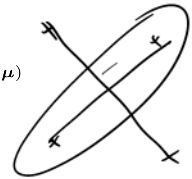
$$var(x) = cov(x, x) = \mathbb{E}[(x - \bar{x})^2]$$

- Standard deviation

$$std(x) = \sqrt{var(x)}$$

The multivariate normal distribution

A distribution for M-dimensional vectors $\boldsymbol{x}$:



$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{M}{2}}\sqrt{|\boldsymbol{\Sigma}|}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$

$$M = 1: \quad \mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$\boldsymbol{\mu}$ is the mean vector and $\boldsymbol{\Sigma}$ is the covariance matrix:

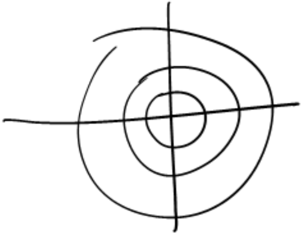
$$\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{x}], \quad \Sigma_{ij} = \mathrm{cov}[x_i, x_j]$$

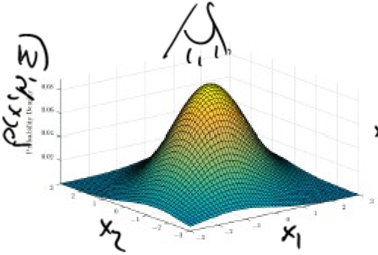
where Σ_{ij} is the covariance between attribute (a random variable) i and attribute j.

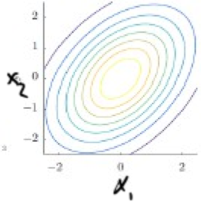
Example: The 2D normal distribution

$$\boldsymbol{\mu} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \mathrm{var}(x_1) & \mathrm{cov}[x_1, x_2] \\ \mathrm{cov}[x_2, x_1] & \mathrm{var}(x_2) \end{bmatrix}$$







The multivariate normal distribution: Covariance

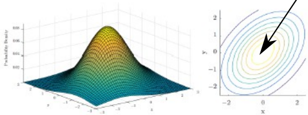
http://www2.imm.dtu.dk/courses/02450/DemoNormal.html

Quiz 3: Covariance

http://www2.imm.dtu.dk/courses/02450/DemoNormal.html

. Match the covariances to the contour plots

$$\mu = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}$$



$$\Sigma = \begin{bmatrix} \mathrm{var}(x_1) & \mathrm{cov}(x_1, x_2) \\ \mathrm{cov}(x_2, x_1) & \mathrm{var}(x_2) \end{bmatrix}$$

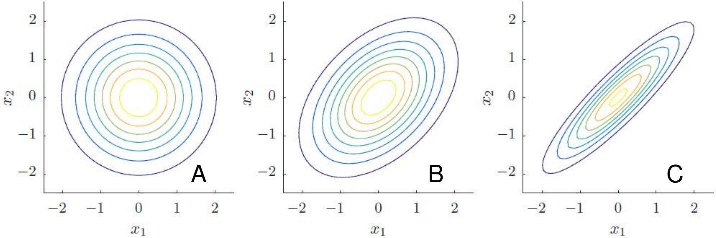
A. Covariance of A is $\mathbf{\Sigma}_A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\mathbf{\Sigma}_C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

B. $\mathbf{\Sigma}_B = \begin{bmatrix} 1 & -0.45 \\ 0.45 & 1 \end{bmatrix}$, $\mathbf{\Sigma}_C = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$

C. $\mathbf{\Sigma}_B = \begin{bmatrix} 10 & 4.5 \\ 4.5 & 10 \end{bmatrix}$, $\mathbf{\Sigma}_C = \begin{bmatrix} 10 & 9 \\ 9 & 10 \end{bmatrix}$

D. $\mathbf{\Sigma}_A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\mathbf{\Sigma}_C = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$

E. Don't know.

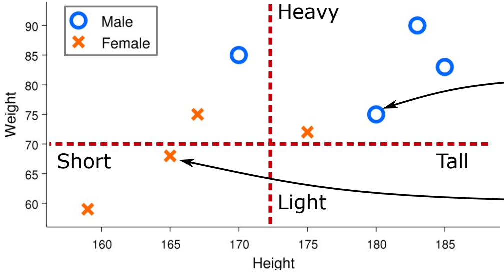


The right answer is D. The covariance has to be positive (because x_1 and x_2 are positively correlated), and the variance is 1 in all cases. Furthermore, since A is axis-aligned, the covariance terms are zero. All in all

$$\boldsymbol{\Sigma}_A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{\Sigma}_B = \begin{bmatrix} 1 & 0.45 \\ 0.45 & 1 \end{bmatrix}, \quad \boldsymbol{\Sigma}_C = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$$

Bayes and Naive Bayes classifiers

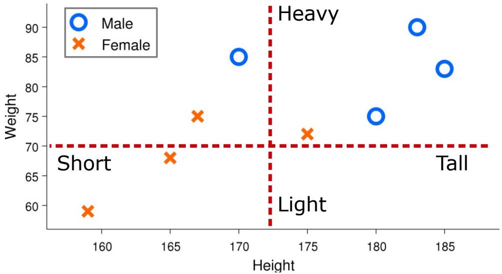
Bayes and Naive-Bayes



Gender y	Tall x_1	Heavy x_2
0	1	1
0	1	1
0	1	1
0	0	1
1	1	1
1	0	1
1	0	0
1	0	0

$$p(y|x_1, x_2) = \frac{p(x_1, x_2|y)p(y)}{\sum_{k=0}^{1} p(x_1, x_2|y = k)p(y = k)}$$

Example 1: Normal Bayes



Gender y	Tall x_1	Heavy x_2
0	1	1
0	1	1
0	1	1
0	0	1
1	1	1
1	0	1
1	0	0
1	0	0

Probability a short, heavy person is male:

$$P(y=0|x_1=0,x_2=1) = \frac{p(x_1=0,x_2=1|y=0)p(y=0)}{\sum_{k=0}^{1} p(x_1=0,x_2=1|y=k)p(y=k)}$$

Example 1: Solution

Probability a short, heavy person is male:

$$P(y=0|x_1=0,x_2=1) = \frac{p(x_1=0,x_2=1|y=0)p(y=0)}{\sum_{k=0}^{1} p(x_1=0,x_2=1|y=k)p(y=k)}$$

$$= \frac{\frac{1}{4}\frac{4}{8}}{\frac{1}{4}\frac{4}{8}+\frac{1}{4}\frac{4}{8}} = \frac{1}{2}$$

A practical problem with Bayesian classifier

- In general:

$$p(y|x_1, x_2, \ldots, x_M) = \frac{p(x_1, x_2, \ldots, x_M|y)p(y)}{\sum_{k=0}^{K-1} p(x_1, x_2, \ldots, x_M|y=k)p(y=k)}$$

$$p(x_1, \ldots, x_M|y=k) = \frac{\text{Nr. obs where } y=k \text{ and we measure } x_1, \ldots, x_M}{\text{Observations where } y=k}$$

A practical problem with Bayesian classifier

- In general:

$$p(y|x_1, x_2, \ldots, x_M) = \frac{p(x_1, x_2, \ldots, x_M|y)p(y)}{\sum_{k=0}^{K-1} p(x_1, x_2, \ldots, x_M|y=k)p(y=k)}$$

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- Naive Bayes assumption

$$p(x_1, x_2, \ldots, x_M|y) = p(x_1|y)p(x_2|y) \times \cdots \times p(x_M|y)$$

A practical problem with Bayesian classifier

- In general:

$$p(y|x_1, x_2, \ldots, x_M) = \frac{p(x_1, x_2, \ldots, x_M|y)p(y)}{\sum_{k=0}^{K-1} p(x_1, x_2, \ldots, x_M|y=k)p(y=k)}$$

$$p(x_1, \ldots, x_M|y=k) = \frac{\text{Nr. obs where } y=k \text{ and we measure } x_1, \ldots, x_M}{\text{Observations where } y=k}$$

- Naive Bayes assumption

$$p(x_1, x_2, \ldots, x_M|y) = p(x_1|y)p(x_2|y) \times \cdots \times p(x_M|y)$$

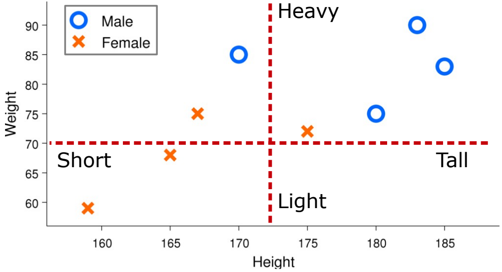
- Naive Bayes classifier

$$\begin{aligned} p(y|x_1, x_2, \ldots, x_M) &= \frac{p(x_1, x_2, \ldots, x_M|y)p(y)}{\sum_{k=0}^{1} p(x_1, x_2, \ldots, x_M|y=k)p(y=k)} \\ &= \frac{p(x_1|y)p(x_2|y) \times \cdots \times p(x_M|y)p(y)}{\sum_{k=0}^{1} p(x_1|y=k)p(x_2|y=k) \times \cdots \times p(x_M|y=k)p(y=k} \end{aligned}$$

Example 2: Solution

- Naive Bayes classifier (Probability someone is a female given they are heavy and tall)

$$p(y=1|x_1=1,x_2=1)=\frac{p(x_1|y)p(x_2|y)p(y)}{\sum_{k=0}^{1}p(x_1|y=k)p(x_2|y=k)p(y=k)}$$



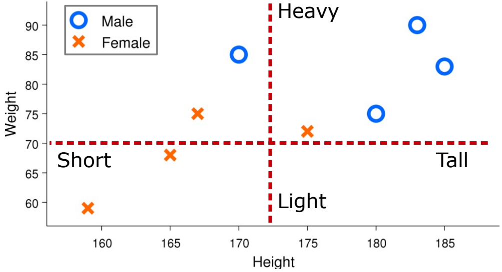
Gender y	Tall x_1	Heavy x_2
0	1	1
0	1	1
0	1	1
0	0	1
1	1	1
1	0	1
1	0	0
1	0	0

Example 2: Solution

- Naive Bayes classifier (Probability someone is a female given they are heavy and tall)

$$p(y=1|x_1=1,x_2=1) = \frac{p(x_1|y)p(x_2|y)p(y)}{\sum_{k=0}^{1} p(x_1|y=k)p(x_2|y=k)p(y=k)}$$

$$= \frac{\frac{1}{4}\frac{2}{4}\frac{1}{2}}{\frac{1}{4}\frac{2}{4}\frac{1}{2}+\frac{3}{4}\frac{4}{4}\frac{1}{2}} = \frac{2}{2+12} = \frac{1}{7}$$



Gender y	Tall x_1	Heavy x_2
0	1	1
0	1	1
0	1	1
0	0	1
1	1	1
1	0	1
1	0	0
1	0	0

Quiz 4: Naive-Bayes (Spring 2012)

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
P1	1	0	0	0	1	1	0	0	1	1
P2	1	0	1	0	0	1	1	1	0	0
P3	0	1	0	1	0	1	0	1	1	1
P4	0	1	1	1	0	0	1	0	0	0
P5	1	0	0	1	1	0	0	1	0	1
P6	1	0	1	1	1	1	1	0	1	0

Table 1: Table indicating whether 10 songs denoted S1–S10 are downloaded to 6 different phones denoted P1–P6. P1 and P2 given in red are phones that belong to females whereas P3, P4, P5, and P6 given in blue belong to males.

The phones P1 and P2 are owned by females whereas P3, P4, P5 and P6 are owned by males (this is indicated in red and blue respectively in Table 1). We would like to predict whether a phone is owned by a male based on whether or not the songs S1, S2 and S3 have been downloaded. We will therefore classify whether the phone belongs to a male or female considering only the attributes S1, S2 and S3 and the data in Table 1. We will apply a Naïve Bayes classifier that assumes independence between these attributes. Given that a phone has installed songs 1, 2 and 3 (i.e., S1=1, S2=1 and S3=1) What is the probability that the phone is owned by a male according to the Naïve Bayes classifier?

A. 1/12

B. 1/6

C. 2/3

D. 1

E. Don't know.

$$p(y|x_1, x_2, \ldots, x_M) = \frac{p(x_1|y) \times \cdots \times p(x_M|y)p(y)}{\sum_{k=0}^{1} p(x_1|y=k) \times \cdots \times p(x_M|y=k)p(y=k)}$$

According to the Naïve Bayes classifier we have

$$P(Male|S1=1,S2=1,S3=1) =$$

$$\frac{\begin{pmatrix} P(S1=1|Male)\times \\ P(S2=1|Male)\times \\ P(S3=1|Male)\times \\ P(Male) \end{pmatrix}}{\begin{pmatrix} P(S1=1|Female)\times \\ P(S2=1|Female)\times \\ P(S3=1|Female)\times \\ P(Female) \end{pmatrix} + \begin{pmatrix} P(S1=1|Male)\times \\ P(S2=1|Male)\times \\ P(S3=1|Male)\times \\ P(Male) \end{pmatrix}}$$

$$= \frac{2/4 \cdot 2/4 \cdot 2/4 \cdot 4/6}{2/2 \cdot 0/2 \cdot 1/2 \cdot 2/6 + 2/4 \cdot 2/4 \cdot 2/4 \cdot 4/6} = 1.$$

Robust estimation and non-binary data

Assume $p(x_1, \ldots, x_M|y) = \prod_{k=1}^{M} p(x_k|y)$

Defining $n_{x_j=k|y=c} = \sum_{i=1}^{N} \delta_{X_{ij},k}\delta_{y,c}$ we have more generally:

Binary case: $$p(x_j = 1|y = c) = \frac{n_{x_j=1|y=c} + \alpha}{N_c + 2\alpha}.$$

Categorical case: $$p(x_j = k|y = c) = \frac{n_{x_j=k|y=c} + \alpha}{N_c + K\alpha}.$$

Continious case: $$p(x_j = x|y = c) = \mathcal{N}(x|\mu = \mu_{j|c}, \sigma^2 = (\sigma_{j|c} + \alpha)^2)$$

$$\mu_{j|c} = \mathbb{E}_{y=c}[x_j] = \frac{1}{N_c}\sum_{i=1}^{N} \delta_{y_i,c} X_{ij},$$

$$\sigma_{j|c} = \hat{\mathrm{std}}_{y=c}[x_j] = \sqrt{\frac{1}{N_c - 1}\sum_{i=1}^{N} \delta_{y_i,c}(X_{ij} - \mu_c)^2}$$

Select these parameters using cross-validation.

Bayesian classification by the multivariate normal distribution

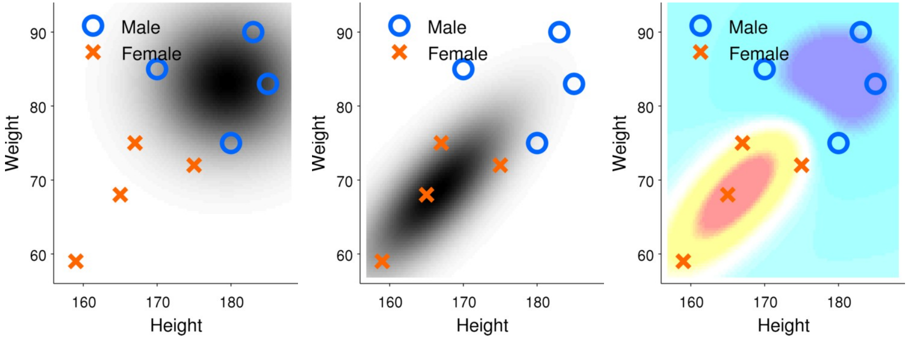
Continuous density estimation

$$P(\boldsymbol{x}|y=c) = \frac{1}{(2\pi)^{\frac{M}{2}} det(\boldsymbol{\Sigma}_c)^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)\right)$$

- Fit a Normal distribution to each class.
  - Compute class mean and covariance

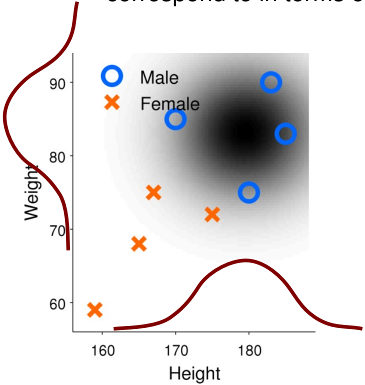
$$P(y=c|\boldsymbol{x}) = \frac{P(\boldsymbol{x}|y=c)P(y=c)}{\sum_{c'} P(\boldsymbol{x}|y=c')P(y=c')}$$

- Classify using Bayes rules as before



Naive Bayes classification with the univariate normal distribution

- What does the Naive Bayes assumption of independence of the attributes correspond to in terms of the parameters of the multivariate normal distribution?

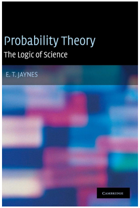


$$p(y=c \mid x_H, x_W) = \frac{p(y=c)\, p(x_H \mid \mu_{H,c}, \sigma_{H,c})\, p(x_W \mid \mu_{H,c}, \sigma_{H,c})}{p(x_H, x_W)}$$

$$p(x_H, x_W) = \sum_{c'} p(y=c')\, \underbrace{p(x_H \mid \mu_{H,c'}, \sigma_{H,c'})}_{\mathcal{N}(x_H \mid \mu=\mu_{H,c},\, \sigma^2=\sigma_{H,c})}\, p(x_W \mid \mu_{H,c'}, \sigma_{H,c'})$$

Summary

Resources



Probability: The logic of science Classical textbook which treats probabilities as states-of-knowledge and discuss many practical and philosophical issues

(https://bayes.wustl.edu/etj/prob/book.pdf)

https://www.khanacademy.org An excellent introduction to probability theory which we recommend as a go-to resource

(https://www.khanacademy.org/math/statistics-probability/probability-library)

http://citeseerx.ist.psu.edu A more in-depth discussion of Bayes in the court room" (http://citeseerx.ist.psu.edu/viewdoc/download;jsessionid=EFE0328140036A4C668BB5B9FC76C9BE?doi=10.1.1.599.8675&rep=rep1&type=pdf)

DTU Demo The multivariate normal distribution

(http://www2.imm.dtu.dk/courses/02450/DemoNormal.html)